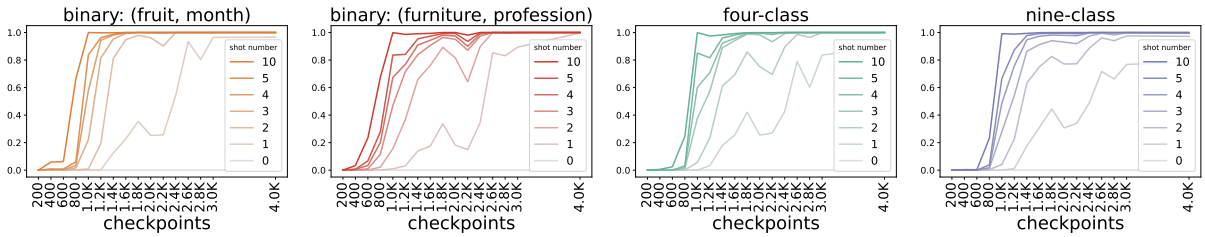

Figure 6: Format accuracy of different tasks at different checkpoints. Each line represents the format accuracy with different numbers of shots (examples) in the prompt.

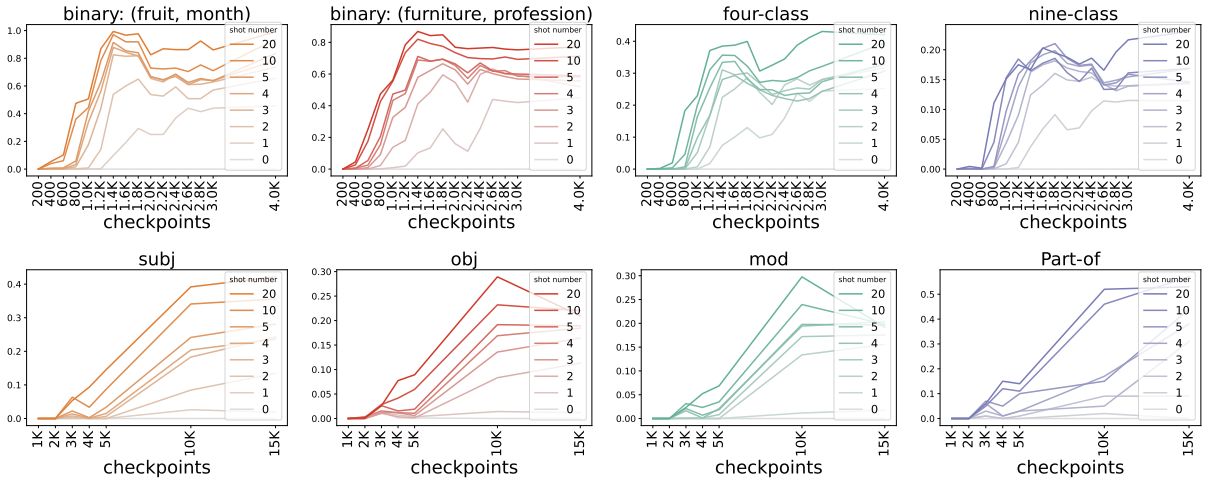

Figure 7: Prediction accuracy of different tasks at different checkpoints. Each line represents the prediction accuracy with different numbers of shots (examples) in the prompt. It is worth noting that these tasks cannot be simply considered as binary classification tasks, which is discussed in Appendix D.

dataset and we set $(i, j) \in \{(50, 50), (50, 450)\}$ to measure the loss reduction at different training checkpoints. Figure 5 shows that the loss difference between later and early tokens decreases quickly from the very beginning of the training. This suggests that from the beginning of the training, the model progressively improves its ability to leverage longer contexts for better predictions.

For format compliance, we construct classification tasks in Table 1. We adopt the few-shot setting of ICL, and the prompt is designed to include several examples followed by a query. Here we ensure that when the number of examples exceeds the number of classes, there is at least one example of each class in the prompt. To test the format compliance ability of the model, we force the model to generate only one token. If the generated token is also a number, matching the format presented in the examples, we consider it to have a correct format. We compute the accuracy of the format to measure the format compliance ability of models.

Figure 6 reports the format accuracy given different numbers of shots at different training checkpoints. For two binary classification tasks, we observe that the model's format accuracy progressively improves as the number of shots increases, starting from the 400th step. This suggests that the model's format compliance ability emerges at the early training stage, and it is independent of the entities involved in the task. For the four-class and nine-class classifications, the model gains improvement with an increasing number of shots at later stages of training (the 600th step and the 800th step, respectively). This indicates that the format compliance to more difficult tasks tends to appear at later training stages. Despite that, the model consistently achieves around 100% format accuracy when using 20 shots at the 1k-th step, and achieves a high accuracy with only one shot after 3k steps. We also measure the format compliance ability from the perspective of the minimum number of shots required to achieve a format accuracy of over 80%. Please refer to Appendix C for details.

For pattern discovery, we still use the above classification tasks, but the difference is that we compute the accuracy of the predicted label. If the model can generate correct labels, we consider it to have successfully discovered and applied the underlying pattern in the prompt. Besides, we also construct four relation justification tasks in Table 1,

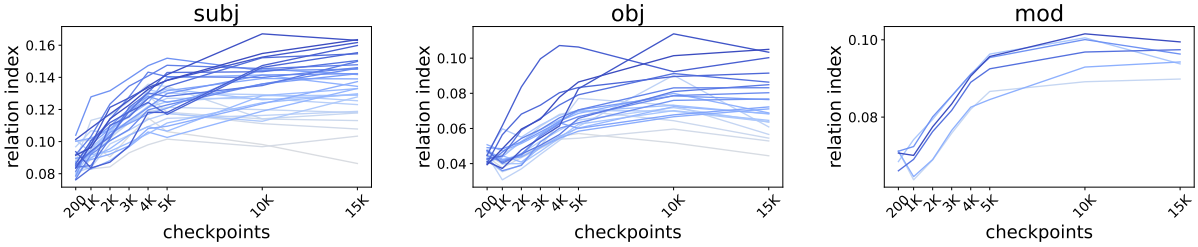

Figure 8: The change curve of average relation indexes of attention heads for syntactic dependency. Each line in the figure represents the relation index of an attention head over training time, and lines are colored according to the value at the 15k-th step.

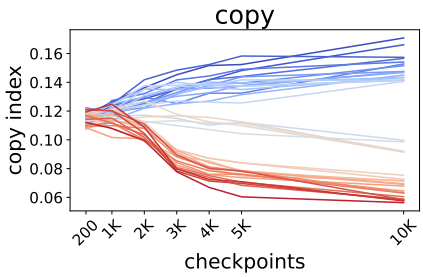

Figure 9: The change curve of copying scores of attention heads. Each line in the figure represents the copying score of an attention head over training time, and lines are colored according to the value at the 10k-th step.

which are related to the syntactic dependency and semantic relationship studied in this paper.

Figure 7 reports the prediction accuracy of the model at different checkpoints. For classification tasks in the first row, the model achieves considerable accuracy with 20 shots at around 1400 steps, indicating that the pattern discovery ability is mastered later than the format compliance. Furthermore, simple binary classification tasks are learned earlier than complex four-class and nine-class classification tasks. The relation justification tasks in the second row, which are more difficult than classification tasks, are learned at later stages, typically starting from around the 2k-th step. After approximately 10k steps, the prediction accuracy tends to saturate. Figure 13 in Appendix E shows that till the end of the training, even with 100 examples, the model cannot fully learn the pattern in the prompt.

Progressive learning of ICL of different levels. From the above results, we can observe a progressive learning process for the different levels of ICL. The loss reduction happens from the beginning of the training, followed by the emergence of format compliance (after 400 steps), and pattern discovery is mastered in the last (after 1k or 2k steps). This discovery aligns with our hypothesis that three levels of ICL have increasing difficulties. Moreover, within the format compliance and pattern discovery, we observe that the model typically learns more challenging tasks at later training stages.

4.2 Correlation Between Semantic Induction Heads and ICL

In this section, we investigate the formation of semantic induction heads during the training process and discover their correlation with ICL. We compute the average relation index of attention heads over all triplets for each syntactic dependency and each semantic relationship in different checkpoints. Here we only ensure $s = \arg\max_{1\le k\le N} A^h_{j,k}$, and do not require $A^h_{j,s}/\max_{k\neq s}\{A^h_{j,k}\} > \tau$ any more, because in the early stage of the training, it is too challenging to find attention heads having an extremely high attention probability on head tokens.

We find that the relation index of some attention heads increases during the same stage as the emergence of the ICL ability. Specifically, Figure 8 shows the change in the relation index for syntactic dependencies of attention heads. We sampled attention heads with an increasing relation index for visualization. It can be observed that relation indexes of some attention heads begin increasing from the beginning of the training, aligning with the emergence of loss reduction. On the other hand, relation indexes of other attention heads begin to increase after around 1k or 2k steps, which coincides with the emergence of the pattern discovery ability. Thus, we infer that the formation of semantic induction heads plays a crucial role in the development of the ICL ability. These semantic induction heads likely contribute to capturing and representing relationships between tokens, which are essential for the ICL ability.

On the other hand, Figure 9 shows the change in copying scores (Olsson et al., 2022; Bansal et al., 2023) of attention heads. Copying scores of some attention heads start increasing from 200 steps, so we infer the copying mechanism is responsible for the loss reduction and the format compliance. It is reasonable because the task used for evaluating

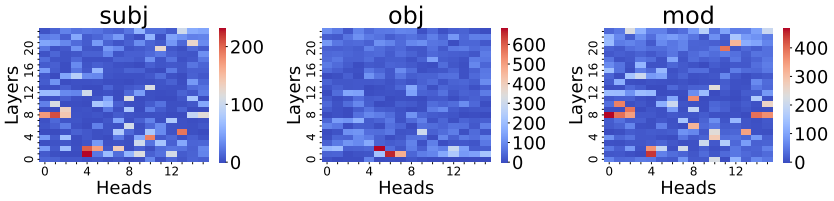

Figure 10: Occurrence of each attention head having the largest value of $\mathbb{E}_j[a_T^{h,j}]$ with $\mathbb{E}_j[a_T^{h,j}] > 0$ over all triplets T of each type of dependency.

the format compliance can be simply achieved by copying the token ("0" or "1") after the colon in the preceding context to the output. More interestingly, copying scores of other heads begin to drop from the 1K step, where the pattern discovery ability emerges. Therefore, we hypothesize that the copying behavior is not always good for ICL, because sometimes direct copying may cause incorrect predictions.

5 More Discussions about Relationships Encoded in Attention Heads

A notable distinction between copying and relationship is that the relationship between tokens depends on the input context, while copying is context-agnostic. Therefore, different inputs may utilize different attention heads. Thus, we propose to find a common group of attention heads that represent specific relationships in different inputs. For each triplet T, we identify the attention head that has the largest value of $\mathbb{E}_j[a_T^{h,j}]$ (larger than 0). Then, we count the occurrence of each head having the largest value among all triplets T for each dependency relationship.

Figure 10 shows the number of occurrences of each attention head having the largest relation index for syntactic dependencies. Please refer to Appendix G for results on semantic relationships. We find that for each type of dependency/relationship, there are around 5∼15 attention heads frequently activated by different inputs. Besides, different relationships tend to share some common attention heads (*e.g.,* layer2, head4 for syntactic dependencies). This may indicate that these human-defined relationships are not mutually exclusive from the LLMs' point of view. In other words, there exists a many-to-many mapping between human-defined relationships and attention heads in LLMs.

6 Conclusion

Previous studies (Elhage et al., 2021; Olsson et al., 2022) in mechanistic interpretability only studied the simple functions in very specific tasks (Wang et al., 2023; Lieberum et al., 2023). In this study, we extended the conventional induction heads to analyze high-level relationships between words/entities in natural languages. Our experiments revealed that specific attention heads encode syntactic dependencies and semantic relationships in natural languages. Furthermore, we identified three levels of the in-context learning ability of LLMs, and experimental results showed they are progressively learned during the training process. Finally, we observed a close correlation between the formation of semantic induction heads and in-context learning ability, strengthening our understanding of in-context learning.

Limitations

Limitations of this paper lie in the following three perspectives. (1) While the proposed relation index has the potential to be adapted to different relationships in various languages, this paper only focuses on syntactic dependency and semantic relations in English. We think it is a promising direction to examine the representation of relationships in different languages and leave it to future work. (2) Although we have extended the simple copying operation to complex semantic relationships, the proposed method is limited to relationships between two tokens/entities. (3) Due to limitations in computational resources, we only conduct experiments on ∼1B models.

Acknowledgement

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